

Neel Maddu

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science | GPA: 4.0

Expected Dec 2027

- **Coursework:** Machine Learning, Artificial Intelligence, Design & Analysis of Algorithms, Data Structures & Algorithms, Systems & Networks, Automata & Complexity, Objects & Design, Applied Combinatorics.
- **U.S. Citizen** | Eligible for security clearance.

EXPERIENCE

Georgia Tech Research Institute (GTRI)

May 2026 – Present

Software Engineer Intern, CIPHER Lab – Threat Analysis Branch

Atlanta, GA

- Build NLP pipelines for entity extraction and semantic analysis across **60K+** unstructured text records and posts, supporting cybersecurity and intelligence analysis workflows in a national security context.
- Evaluate **15 agentic configurations** against **6 categories** of adversarial and injection-style attacks across an isolated **3-VM** testbed, quantifying how prompt design, tool permissions, and guardrail placement drive failure rates.
- Brief **5 program stakeholders** on model-vulnerability findings, informing security assessments and program-level architecture decisions.

Georgia Institute of Technology – VIP Program

Jan. 2026 – Present

Undergraduate Researcher, Automated Algorithm Design

Atlanta, GA

- Own the judge for an automated LLM jailbreaking pipeline built on LLM-Guided Evolution (LLM-GE), cited as prior work by Google DeepMind’s AlphaEvolve; designed the scoring formula the team adopted as the search’s fitness function.
- Proposed a two-judge setup with HarmBench as an independent reporting judge to catch the search overfitting its own scorer; first runs on the PACE ICE HPC cluster showed a **46.2% vs. 30.6%** attack-success gap between judges.

Georgia Tech iOS Club

Aug. 2025 – Present

Tech Lead (Aug. 2026 – Present); previously Senior iOS Developer and iOS Developer

Atlanta, GA

- Co-lead a **20+ developer** team building Contour, an on-device iOS accessibility app that guides blind and low-vision users to buttons on unfamiliar appliance panels using Vision and haptic/audio feedback.
- Restructured the shared Swift codebase for **3 sub-teams** and **12 parallel lanes** working against common interface contracts; maintain CI and tooling, including the migration to Xcode 27 and iOS 27.
- Grew from developer to Tech Lead across 3 apps in 3 semesters, including Slack API messaging for SmartCompose; authored **17** and merged **22** pull requests across the club’s repos.

PROJECTS

MARTA Live Transit Tracker | Python, FastAPI, SQLite, SwiftUI, WidgetKit, GTFS-Realtime

2026

- Built a full-stack transit tracker: a 24/7 FastAPI/SQLite backend polling MARTA’s GTFS-Realtime and rail feeds, accumulating **575K+ arrival observations** across 78 routes and 6,200+ stops, feeding a native SwiftUI app tracking ~200 live vehicles at 15-second refresh.
- Resolved feed defects absent from official docs – reconstructed missing bus delays via trip-to-schedule joins against **2.4M GTFS rows** and hand-wrote a protobuf decoder validated byte-for-byte against the reference parser – and integrated OpenTripPlanner for delay-aware trip planning, backed by **59 automated tests**.

Hidden Quakes (HackGT 13) | Python, PyTorch, SeisBench, PyOcto, ObsPy, React Three Fiber

2026

- Owned the seismology lane of a 4-person, 36-hour build that rebuilt a **15x denser** earthquake catalog for Utah’s FORGE geothermal site from raw public waveforms: **654 candidate events vs. 43** public, with **43/43** catalog recall.
- Built PyOcto event association, grid-search relocation with station statics, quality tiering, and magnitude calibration (leave-one-out MAE 0.11) over 27K picks from 24 stations; a 20-shuffle null test found **0** strict-tier chance events.

LLM Red-Teaming Evaluation Pipeline | Python, Streamlit, Ollama, Anthropic API, pytest

2026

- Built an extensible prompt-injection evaluation framework (pluggable provider/target/classifier/defense layers) testing local and frontier LLMs against **46 attack payloads** across 4 categories and 41 techniques, with 35 tests and CI.
- Engineered a dual rule-based + LLM-as-judge scoring pipeline and Streamlit dashboard quantifying a robustness gap of up to **47% breakthrough** on local 8B models versus near-zero on a frontier model.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Swift, SQL

ML & AI: PyTorch, TensorFlow, scikit-learn, FAISS, MediaPipe, NLP, LLM Evaluation, Graph Analytics

Frameworks & Libraries: FastAPI, Node.js, React, Next.js, SwiftUI, Core ML, Vision, Streamlit, pandas, NumPy

Data & Infrastructure: MySQL, MongoDB, SQLite, Firebase, Docker, Git, GitHub Actions, Jenkins, CI/CD